SOLVING RATIONAL AND RADICAL EQUATIONS

BASIC ALGEBRA

prepared by:

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Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Solving Rational Equations

Solving Radical Equations

SOLVING RATIONAL EQUATIONS

Solve $\frac{x}{x+3} = 1 + \frac{1}{x}$

LCD ,

$x(x+3)$

$$x(x+3)\left(\frac{x}{x+3} = 1 + \frac{1}{x}\right)$$

Solve $\frac{x}{x+3} = 1 + \frac{1}{x}$

LCD ,

$x(x+3)$

$x(x+3)\left(\frac{x}{x+3} = 1 + \frac{1}{x}\right)$

$x^2 = x(x+3) + (x+3)$

$x^2 = x^2 + 3x + x + 3$

$0 = 4x + 3$

$4x = -3$

$\boxed{x = -\frac{3}{4}}$

ans

Solve $\frac{4c}{c-5} - \frac{1}{c+1} = \frac{3c^2+3}{c^2-4c-5}$

LCD ,

$c^2 - 4c - 5 = (c-5)(c+1)$

$$(c-5)(c+1)\left(\frac{4c}{c-5} - \frac{1}{c+1} = \frac{3c^2+3}{c^2-4c-5}\right)$$

Solve $\frac{4c}{c-5} - \frac{1}{c+1} = \frac{3c^2+3}{c^2-4c-5}$

LCD ,

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Solve $\frac{4c}{c-5} - \frac{1}{c+1} = \frac{3c^2+3}{c^2-4c-5}$

LCD ,

$c^2 - 4c - 5 = (c-5)(c+1)$

$(c-5)(c+1)\left(\frac{4c}{c-5} - \frac{1}{c+1} = \frac{3c^2+3}{(c-5)(c+1)}\right)$

$4c(c+1) - (c-5) = 3c^2 + 3$

$4c^2 + 4c - c + 5 = 3c^2 + 3$

$c^2 + 3c + 2 = 0$

$(c+1)(c+2) = 0$

$(c+1) = 0$ $\quad$ $(c+2) = 0$

$c = -1$ $\quad$ $\boxed{c = -2}$

ans

SOLVING RADICAL EQUATIONS

Find all the real solutions for the equation

$$x + \sqrt{x} = 12$$

$$\sqrt{x} = 12 - x$$

$$x = (12 - x)^2$$

$$x = 144 - 24x + x^2$$

Find all the real solutions for the equation

$$x + \sqrt{x} = 12$$

$$\sqrt{x} = 12 - x$$

$$x = (12 - x)^2$$

$$x = 144 - 24x + x^2$$

$$x^2 + 25x - 144 = 0$$

-144

-9 $\quad$ -16

25

$$(x - 9)(x - 16) = 0$$

$$(x - 9) = 0 \qquad (x - 16) = 0$$

$$x = 9 \qquad x = 16$$

Check ,

$$9 + \sqrt{9} = 12 \qquad 16 + \sqrt{16} = 12$$

$$9 + 3 = 12 \qquad 16 + 4 \neq 12$$

$$\boxed{x = 9}$$

ans

Find all the **real solutions** for the equation

$$2p^{4/5} = \frac{1}{8}$$

Find all the **real solutions** for the equation

$$\frac{1}{2}\left(2p^{4/5} = \frac{1}{8}\right)$$

$$p^{\frac{4}{5}} = \frac{1}{16}$$

$$p^{\left(\frac{4}{5}\right)\frac{5}{4}} = \left(\frac{1}{16}\right)^{\frac{5}{4}}$$

Find all the **real solutions** for the equation

$$\frac{1}{2}\left(2p^{4/5} = \frac{1}{8}\right)$$

$$p^{\frac{4}{5}} = \frac{1}{16}$$

$$p^{1} = \left(\frac{1}{16}\right)^{\frac{5}{4}}$$

Find all the **real solutions** for the equation

$$\frac{1}{2}\left(2p^{4/5} = \frac{1}{8}\right)$$

$$p^{\frac{4}{5}} = \frac{1}{16}$$

$$p^{1} = \frac{1^{\frac{5}{4}}}{16^{\frac{5}{4}}}$$

Find all the **real solutions** for the equation

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$$p^{1} = \frac{1}{16^{\frac{5}{4}}}$$

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$$p^{1} = \frac{1}{16^{\frac{5}{4}}}$$

Find all the **real solutions** for the equation

$$\frac{1}{2}\left(2p^{4/5} = \frac{1}{8}\right)$$

$$p^{\frac{4}{5}} = \frac{1}{16}$$

$$p^{1} = \frac{1}{(2^{4})^{\frac{5}{4}}}$$

Find all the **real solutions** for the equation

$$\frac{1}{2}\left(2p^{4/5} = \frac{1}{8}\right)$$

$$p^{\frac{4}{5}} = \frac{1}{16}$$

$$p^{1} = \frac{1}{2^{5}}$$

Find all the **real solutions** for the equation

$$\frac{1}{2}\left(2p^{4/5} = \frac{1}{8}\right)$$

$$p^{\frac{4}{5}} = \frac{1}{16}$$

$$p = \frac{1}{32}$$

Check ,

$$2\left(\frac{1}{32}\right)^{4/5} = \frac{1}{8}$$

$$2\left(-\frac{1}{32}\right)^{4/5} = \frac{1}{8}$$

Find all the **real solutions** for the equation

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$$p^{\frac{4}{5}} = \frac{1}{16}$$

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Check ,

$$2\left(\frac{1^{4/5}}{32^{4/5}}\right) = \frac{1}{8}$$

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Find all the **real solutions** for the equation

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$$2\left(\frac{-1^{4/5}}{32^{4/5}}\right) = \frac{1}{8}$$

$$2\left(\frac{\left(\sqrt[5]{1}\right)^4}{\left(\sqrt[5]{32}\right)^4}\right) = \frac{1}{8}$$

$$2\left(\frac{\left(\sqrt[5]{-1}\right)^4}{\left(\sqrt[5]{32}\right)^4}\right) = \frac{1}{8}$$

Find all the **real solutions** for the equation

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$$p^{\frac{4}{5}} = \frac{1}{16}$$

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Check ,

$$2\left(\frac{1^{4/5}}{32^{4/5}}\right) = \frac{1}{8}$$

$$2\left(\frac{-1^{4/5}}{32^{4/5}}\right) = \frac{1}{8}$$

$$2\left(\frac{(1)^4}{\left(\sqrt[5]{32}\right)^4}\right) = \frac{1}{8}$$

$$2\left(\frac{(-1)^4}{\left(\sqrt[5]{32}\right)^4}\right) = \frac{1}{8}$$

Find all the **real solutions** for the equation

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$$2\left(\frac{-1^{4/5}}{32^{4/5}}\right) = \frac{1}{8}$$

$$2\left(\frac{(1)^4}{(2)^4}\right) = \frac{1}{8}$$

$$2\left(\frac{(-1)^4}{(2)^4}\right) = \frac{1}{8}$$

Find all the **real solutions** for the equation

$$\frac{1}{2}\left(2p^{4/5} = \frac{1}{8}\right)$$

$$p^{\frac{4}{5}} = \frac{1}{16}$$

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$$2\left(\frac{-1^{4/5}}{32^{4/5}}\right) = \frac{1}{8}$$

$$2\left(\frac{1}{16}\right) = \frac{1}{8}$$

Find all the **real solutions** for the equation

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$$2\left(\frac{1^{4/5}}{32^{4/5}}\right) = \frac{1}{8}$$

$$\frac{1}{8} = \frac{1}{8}$$

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Find all the **real solutions** for the equation

$$\frac{1}{2}\left(2p^{4/5} = \frac{1}{8}\right)$$

$$p^{\frac{4}{5}} = \frac{1}{16}$$

$$\boxed{p = \pm\frac{1}{32}}$$

ans

Check ,

$$2\left(\frac{1^{4/5}}{32^{4/5}}\right) = \frac{1}{8}$$

$$\frac{1}{8} = \frac{1}{8}$$

$$2\left(\frac{-1^{4/5}}{32^{4/5}}\right) = \frac{1}{8}$$

$$\frac{1}{8} = \frac{1}{8}$$

SEATWORK